

The Analysis of the Performance of Data Analysis Students

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2026 Automotive Trends: Gas vs. Electric Vehicle Study

1. Introduction

The automotive industry is currently at a pivotal crossroads as it transitions from traditional Internal Combustion Engines (ICE) to Electric Vehicles (EV). As we approach 2026, consumer preferences are being shaped by environmental concerns, fluctuating fuel prices, and rapid advancements in battery technology. This report explores consumer perceptions and the primary factors influencing the choice between gas-powered and electric vehicles.

2. Research Question

What are the primary factors influencing consumer preference between Gas-Powered Vehicles and Electric Vehicles (EVs) in 2026?

3. Hypothesis

- **Null Hypothesis (H_0):** Consumer vehicle preference (Gas vs. Electric) is independent of their primary concern (e.g., Price, Environment, or Infrastructure).
- **Alternative Hypothesis (H_1):** Consumer vehicle preference is dependent on their primary concern, suggesting that specific priorities significantly drive the choice of vehicle type.

Population of Interest: Potential car buyers and current vehicle owners.

- **Sampling Method:** The survey used a convenience sampling method by sharing the form link in CIS groups for all four levels at EU. This approach was chosen for its accessibility and quick data collection.

- **Bias Identification:**

To ensure objectivity, the survey used neutral terminology and avoided "pro-green" or "properformance" leading questions, providing a balanced set of concerns and benefits for both vehicle types.

4. Survey Questions:

- **What is your gender**
- **What is your major**
- **What is your age**
- **What is your current academic year**
- **Do you currently own or regularly drive a vehicle**
- **What is your primary vehicle**
- **What is your average daily distance traveled in kilometers**
- **How often do you drive more than 150 Km in a single day? (Daily, Weekly, Monthly, 1–2 times a year, Never**
- **What is your primary parking situation at home**
- **What do you believe is the minimum range (in km) an EV needs to be "practical" for daily use?**
- **Estimate the annual maintenance cost (oil, brakes, engine, etc.) for a Gas car.(in EGP)**
- **Estimate the annual maintenance cost for an Electric car. (in EGP)**
- **If you were buying a car today, what is the #1 barrier stopping you from going Electric**
- **On a scale of 1–10, how much do you trust public charging infrastructure to be working when you arrive? (Linear Scale: 1 = Not at all, 10 = Completely) • Will your next vehicle purchase likely be Electric**

Online survey link:

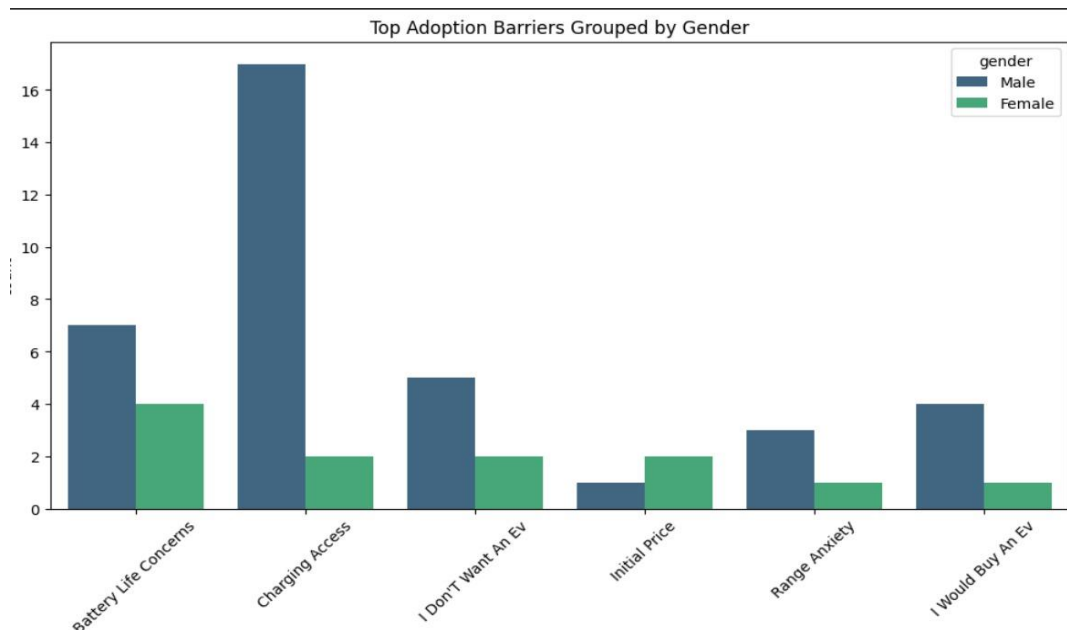
<https://docs.google.com/forms/d/e/1FAIpQLScMyUlJyOjhXwGv97b3BwZtUw2TAtdWMogDPBcwreySh7czg/viewform?pli=1>

Number of samples collected:49

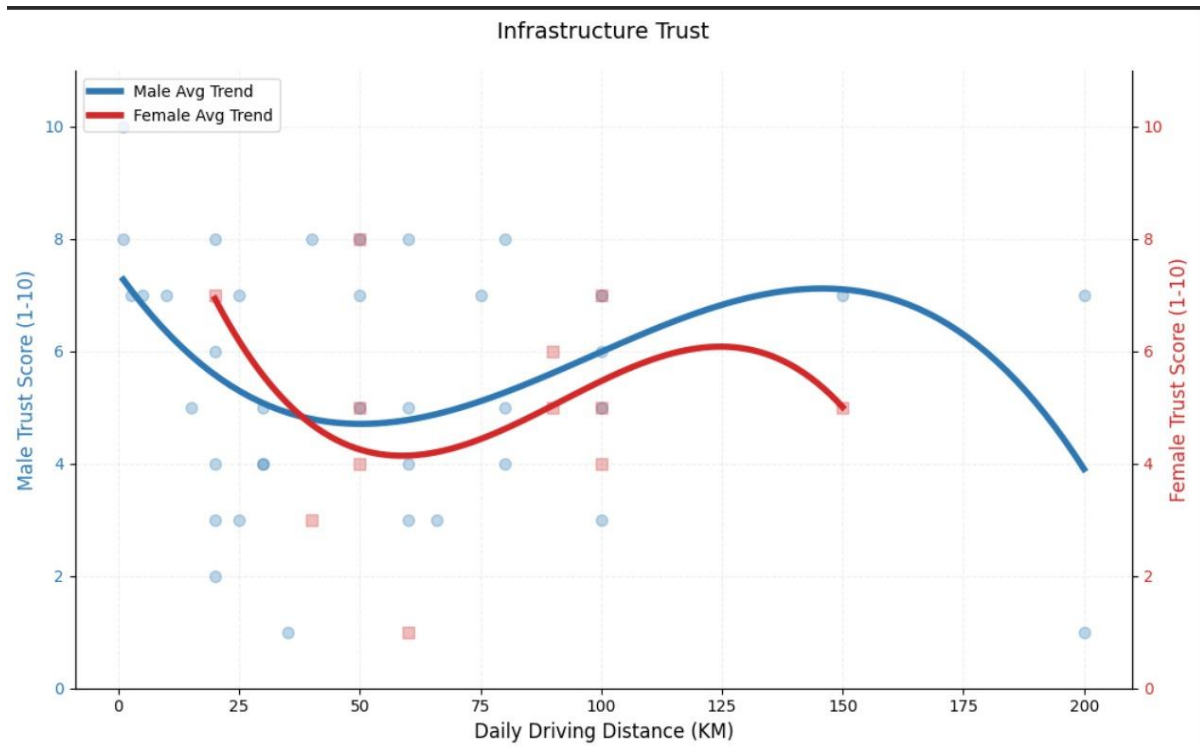
5. Analysis:

To analyze the data collected from this survey, we calculated descriptive statistics and used bar charts to visualize patterns and trends in the responses.

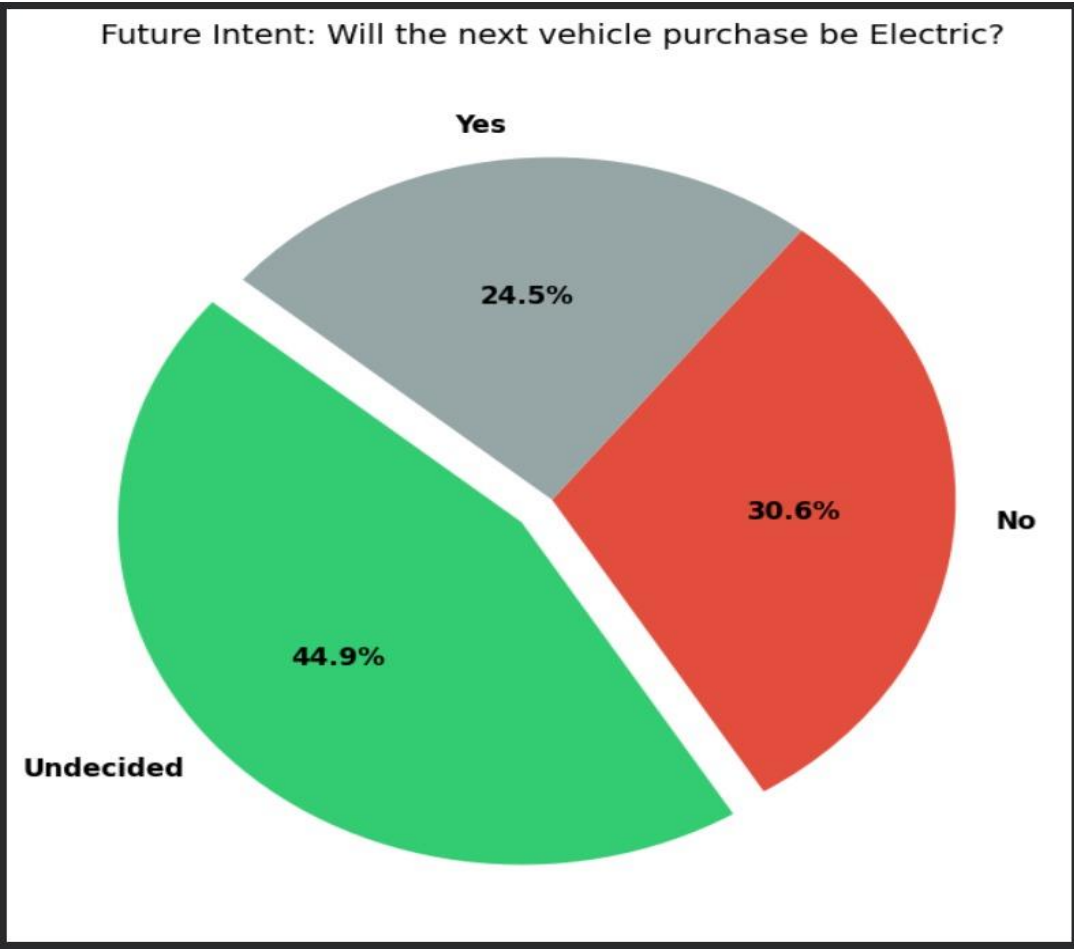
SUMMARY: Central Tendencies					
	daily_dist_km	min_range_req	est_gas_maint	est_ev_maint	infra_trust
mean	60.62	362.24	30014.80	26226.43	5.35
median	50.00	300.00	20000.00	10000.00	5.00
mode	100.00	300.00	20000.00	5000.00	7.00



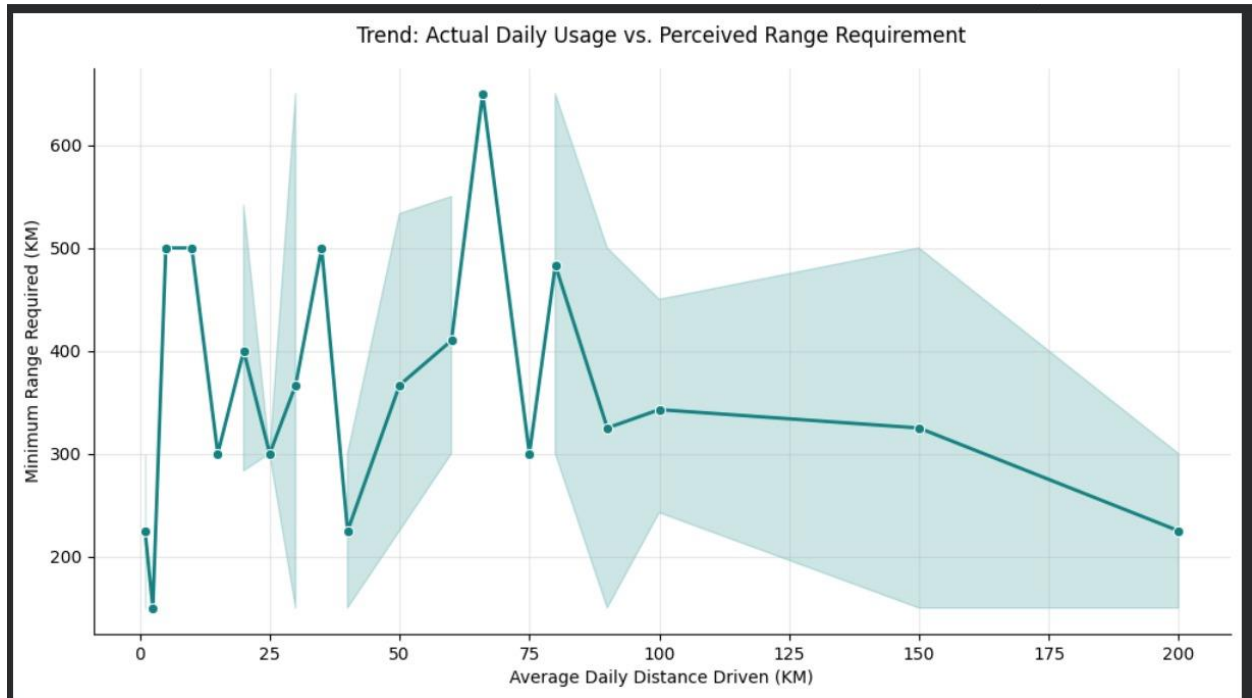
This chart illustrates the main barriers to EV adoption. The initial price and charging access are the most common concerns, showing that financial and infrastructure challenges are key obstacles.



This chart shows the level of trust in charging infrastructure. Most responses fall in the mid to low range, indicating limited confidence in public charging stations.

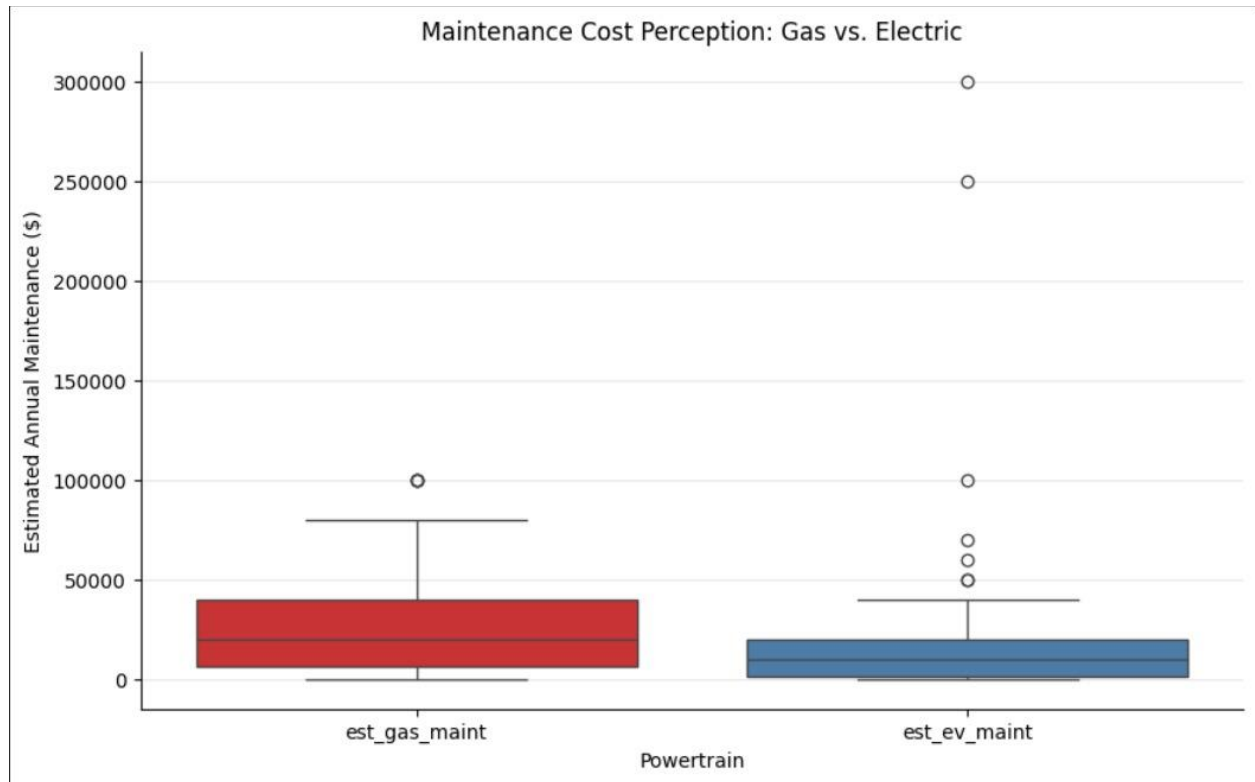


This chart represents future EV purchase intentions. A large portion of respondents are undecided, suggesting uncertainty about switching to electric vehicles.

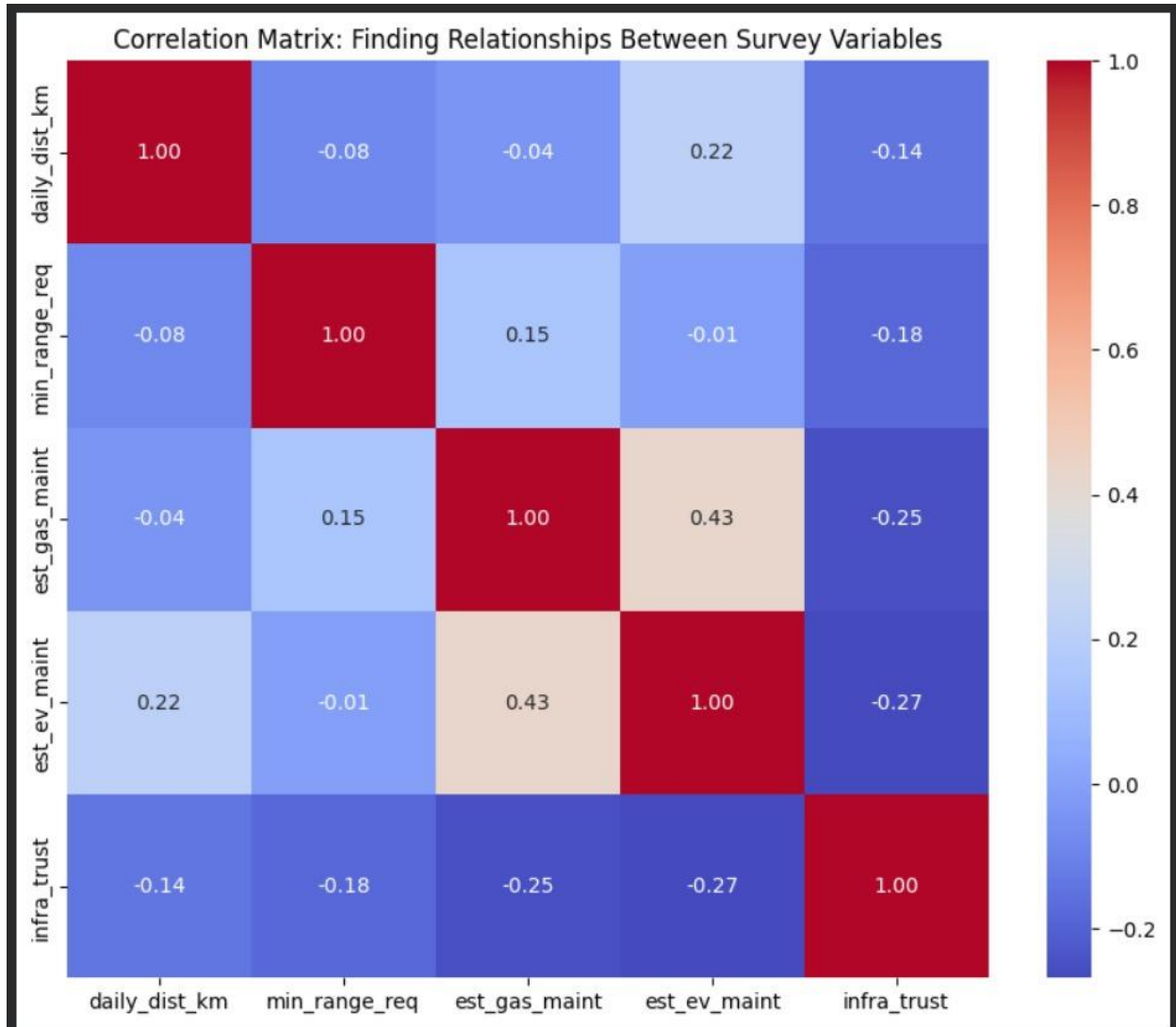


This graph illustrates the relationship between the **average daily distance driven** and the **perceived minimum range requirement** for electric vehicles.

It shows that as daily driving distance increases, the required range generally increases as well; however, there is noticeable variability, indicating that some users prefer a higher range than their actual needs, likely due to range anxiety.



The graph compares the distribution of estimated annual maintenance costs for gas-powered versus electric vehicles, showing that gas cars generally have higher typical costs while electric cars exhibit more extreme high-cost outliers.



The graph shows the correlation matrix between survey variables, indicating how factors like daily distance, range requirements, maintenance cost estimates, and infrastructure trust are positively or negatively related to each other.

6. Any potential issues

Sampling Bias: The survey mainly targets students, which may not represent the general population.

Gender Imbalance: Unequal gender representation may influence the results.

Lack of EV Experience: Most respondents do not own EVs, so opinions may be based on perception.

